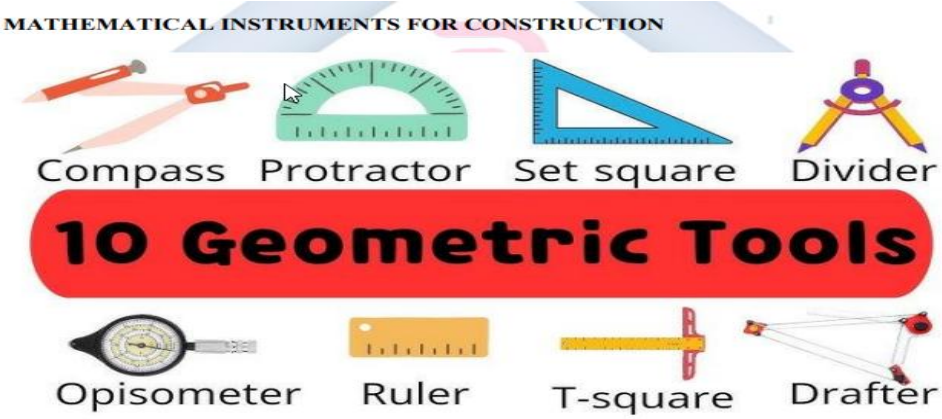
JSS 1 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 4: CONSTRUCTIONS

MATHEMATICAL INSTRUMENTS FOR CONSTRUCTION



1. Ruler (Straight Edge)

- Used to draw straight lines
- Used to measure lengths
- Marked in centimeters (cm) and millimeters (mm)

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2. Compass (Pair of Compasses)

- Used to draw circles and arcs
- Used to transfer measurements
- Has two legs:
    - Needle point
    - Pencil

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3. Protractor

- Used to measure angles
- Used to draw angles
- Marked from 0° to 180°

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4. Set Squares

- Triangular instruments
- Used to draw perpendicular and parallel lines

- Common angles:
  - 30°, 60°, 90°
  - 45°, 45°, 90°

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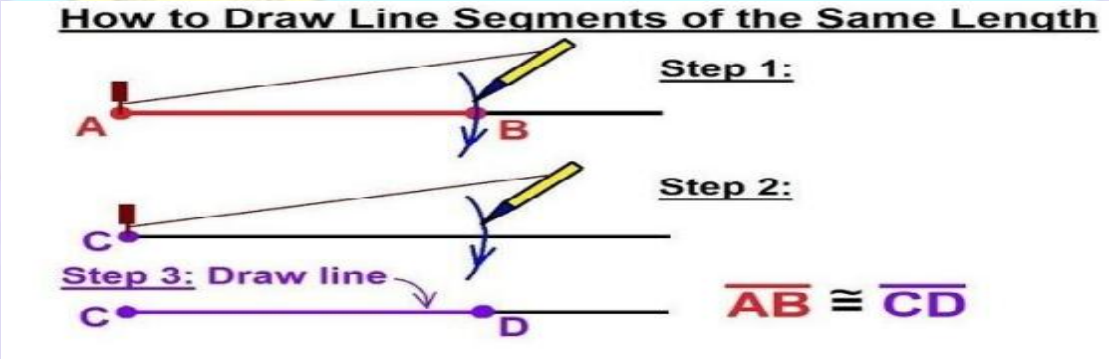
5. Pencil

- Use a sharp pencil (HB or H)
- Ensures accuracy in construction

BASIC CONSTRUCTION TECHNIQUES

CONSTRUCTION 1: DRAWING A LINE SEGMENT

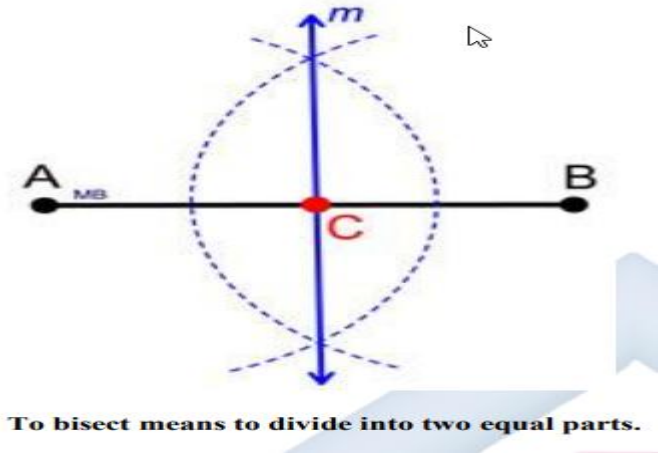


To draw AB = 6 cm

Steps:

1. Mark point A
2. Place ruler with 0 cm at A
3. Mark point B at 6 cm
4. Join A to B
5. Label AB = 6 cm

CONSTRUCTION 2: BISECTING A LINE SEGMENT



To divide a line into two equal parts

Steps:

1. Draw line AB
2. With center A and radius greater than half AB, draw arcs above and below
3. With center B and same radius, draw arcs cutting previous arcs
4. Mark intersection points P and Q
5. Join P and Q
6. Line PQ meets AB at M (midpoint)

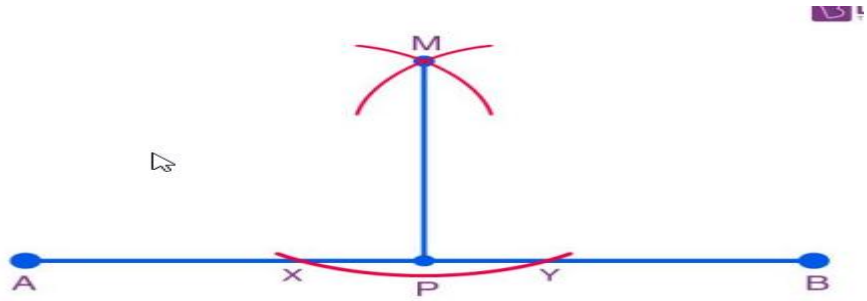
Result:
AM = MB

Note:
PQ is perpendicular to AB

Example

Line = 8 cm → Each part = **4 cm**

CONSTRUCTION 3: PERPENDICULAR LINES



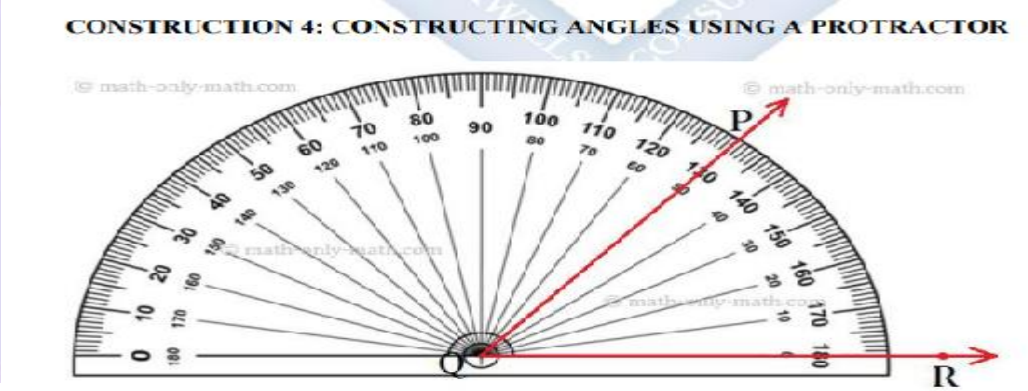
Method 1: Using Set Square

1. Draw line AB
2. Mark point P
3. Place set square along AB
4. Slide until edge passes through P
5. Draw perpendicular line

Method 2: Using Compass (At a Point on Line)

1. Draw AB
2. With center B, draw arc cutting AB
3. Use intersections to construct arcs
4. Join final point to B
5. Result is perpendicular

CONSTRUCTION 4: CONSTRUCTING ANGLES WITH PROTRACTOR



Example: 60°

Steps:

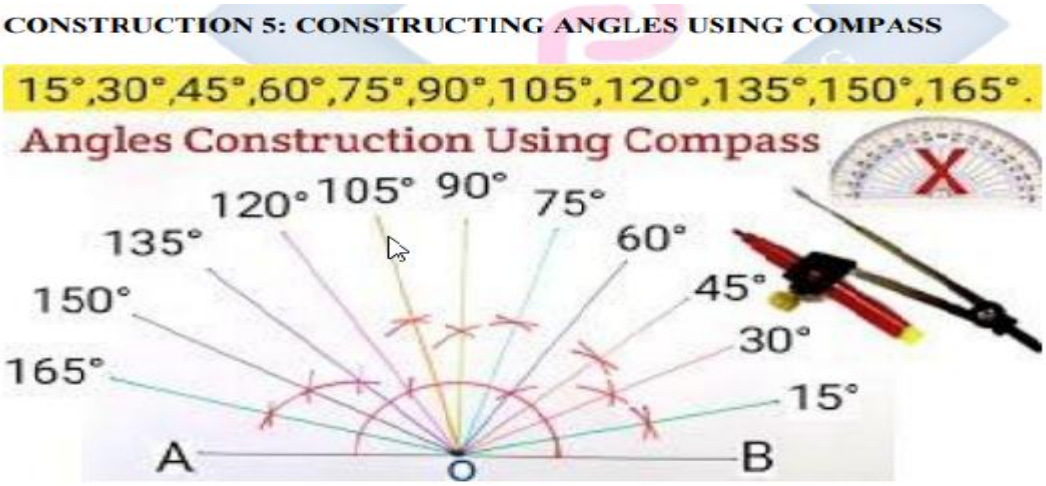
1. Draw line AB
2. Place protractor at A
3. Mark 60°
4. Join A to mark

Result: ∠BAC = 60°

Common Angles

30°, 45°, 60°, 90°, 120°, 135°, 150°

CONSTRUCTION 5: CONSTRUCTING ANGLES WITH COMPASS



To Construct 60°

1. Draw line AB
2. With center A, draw arc cutting AB
3. With center at cut point, draw another arc
4. Join A to intersection

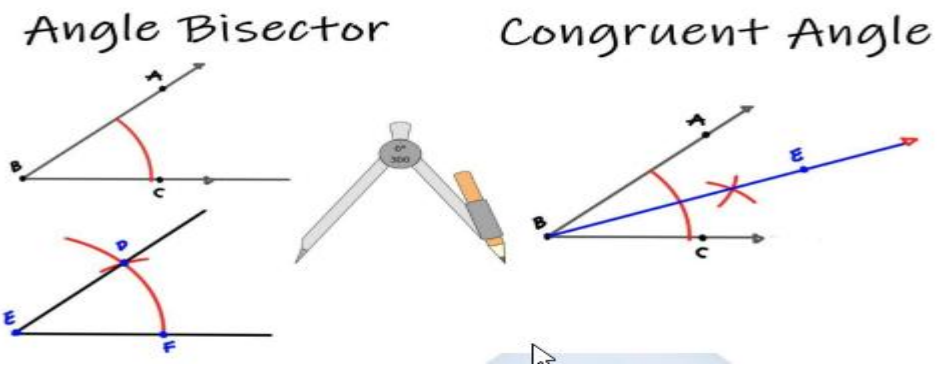
To Construct 90°

1. Draw AB
2. Construct arcs step by step
3. Join final intersection

To Construct 120°

1. Construct 60°
2. Extend arc
3. Join to form 120°

CONSTRUCTION 6: BISECTING AN ANGLE



Steps:

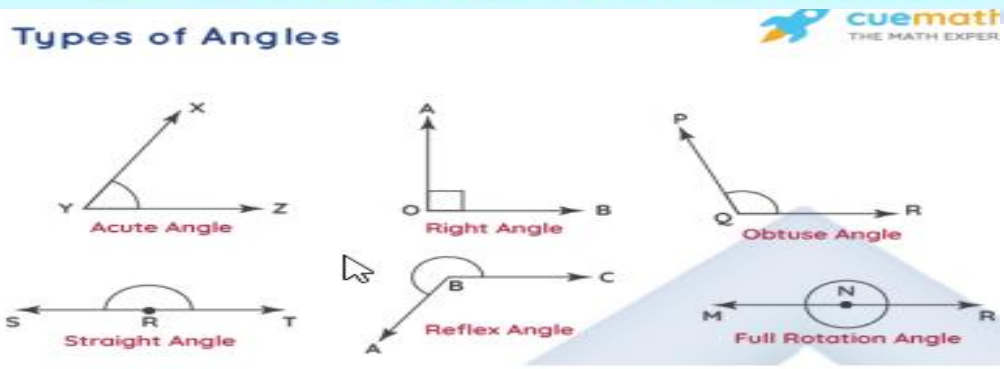
1. Draw angle ABC
2. Draw arc cutting both arms
3. From cut points, draw intersecting arcs
4. Join vertex to intersection

Result:
Angle is divided into two equal parts

Examples

- 80° → 40° each
- 90° → 45° (by bisection)

CONSTRUCTION 7: SPECIAL ANGLES



30°

- Construct 60°
- Bisect it

45°

- Construct 90°
- Bisect it

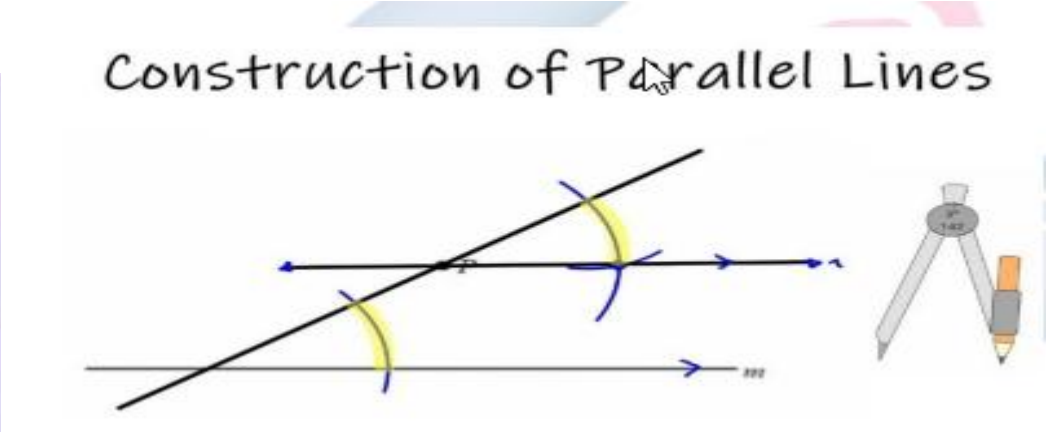
15°

- Construct 30°
- Bisect it

75°

- Combine 60° + 15°

CONSTRUCTION 8: PARALLEL LINES



Method 1: Using Set Squares

1. Draw AB
2. Place set squares
3. Slide and draw line
4. New line is parallel

Method 2: Using Compass

1. Draw line through point P
2. Copy corresponding angle
3. Form parallel line

CONSTRUCTION TIPS

- Keep pencil sharp
- Draw lightly

- Do not erase arcs
- Label clearly
- Keep compass firm
- Practice regularly

PRACTICAL APPLICATIONS

- Architecture (building design)
- Engineering (technical drawings)
- Art (patterns and symmetry)
- Carpentry (accurate measurements)
- Navigation (maps and direction)

CLASSWORK

1. Name three construction instruments
2. Explain how to bisect a line
3. How do you construct 90° using a compass?
4. What is the first step in angle bisection?
5. How do you construct 45°?

ASSIGNMENT

1. Draw and bisect a 7 cm line
2. Construct 60° using a compass
3. Bisect 100° → Find each angle
4. Describe steps to construct a perpendicular line
5. Explain two methods of constructing 30°